

# From Particles to Waves

Towards Coupling Dynamic  
Snow Modelling and Spectral  
Elements to Detect  
Avalanche Breakoff in DAS  
Data

**Salome Bachmann**

August 6, 2026



# Overview

- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations
- **Goal:** Couple dynamic snow modelling with SE to model crack propagation in snow

# Nomenclature & Abbreviations

- **SE, FE, FEM**: Spectral elements, Finite Element (Method), done in SALVUS
- **MPM**: Material Point Method Numerical Modeling
- **PST**: Propagation Saw Test
- **F-k**: Frequency-Wavenumber Plot
- **DAS**: Distributed Acoustic Sensing (Fibre-Optics)
- **STF**: Source-time function (the wavelet fired by SALVUS)

## Why model crack propagation in snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weak layer (Gaume et al., 2017)

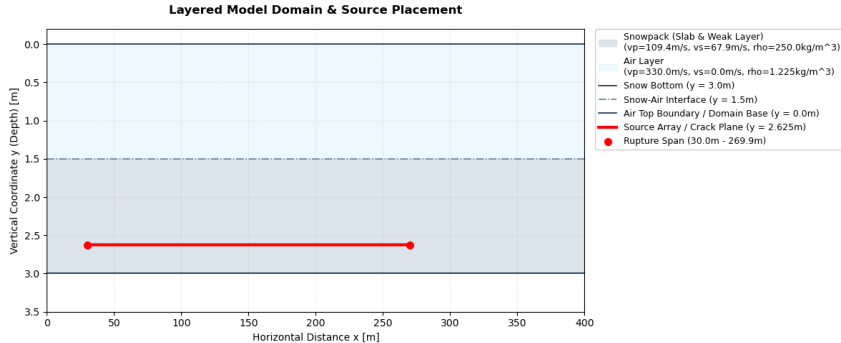
# Relevance

- Crack propagation at different speeds: sub-Rayleigh (below  $v_s$  of material) and Supershear (above  $v_s$ ) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation  $\rightarrow$  critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

# Research Questions

- What FE framework is required to to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

# Simple Model I



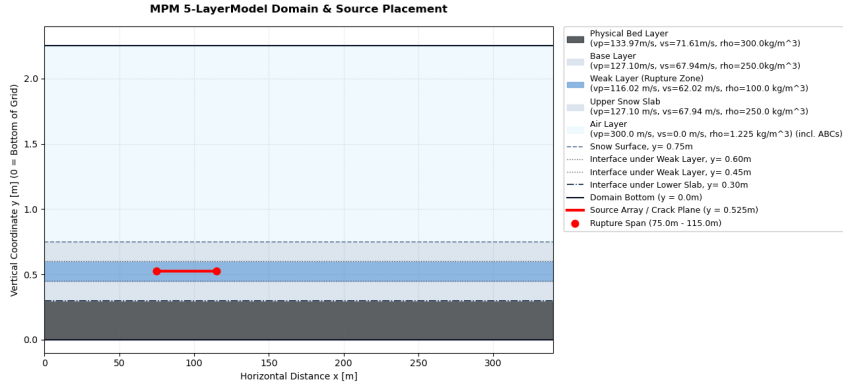
**Figure:** Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

## Simple Model II

- Set up to test custom STF in SALVUS, see physical effects on output
- Sources spaced over 240m in 10cm intervals
- 20Hz central frequency Ricker wavelet
- Source Wavelet in xx (1), xy (1) and yy (-1) direction
- Time delay of wavelet calculated based on propagation speed: sub-Rayleigh speed is  $0.6v_s$ , Supershear  $1.6v_s$  (Trottet et al., 2022)



# MPM-Guided Model I



**Figure:** Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

## MPM-Guided Model II

- Source-time functions obtained from MPM-simulation of PST by Trottet et al., 2022
- Source Wavelet in all three directions, but irregular and different for every source point

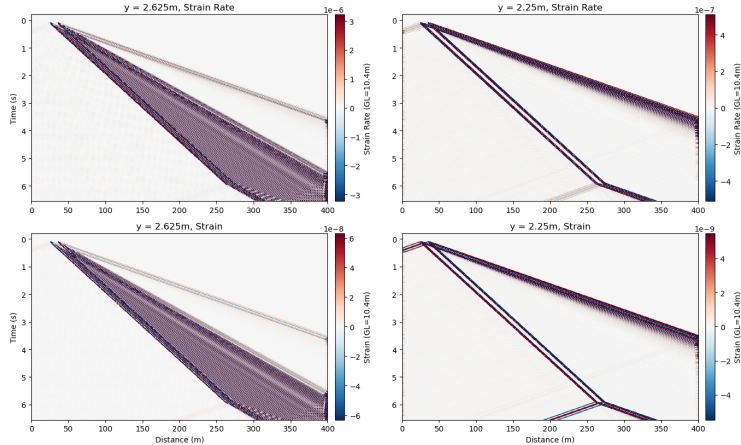
# Results

- SALVUS Output: Displacement field  $u$
- $\frac{\partial u_i}{\partial t} \rightarrow$  velocity in  $x$  and  $y$
- $\frac{\partial u_i}{\partial j} \rightarrow$  strain in  $x$  and  $y$   $\epsilon$
- $\frac{\partial^2 u_i}{\partial t \partial t} \rightarrow$  strain rate  $\dot{\epsilon}$
- Simulation of DAS data from  $\epsilon$ ,  $\dot{\epsilon}$  using Pyber Gauge length class by **noeFrontiersFullwaveformSeismology2025**, gauge length according to Turquet et al., 2024
- Spate-time plots of strain, strain rate and velocity at different receiver depths
- Frequency-wavenumber plots with theoretical speeds and dispersion curves
- Animations of velocity wavefields

## Simple Sub-Rayleigh: Strain and Strain Rate I

# Simple Sub-Rayleigh: Strain and Strain Rate II

Sub-Rayleigh: Simulated DAS Strain Rate and Strain at  $y = 2.625\text{m}$  and  $y = 2.25\text{m}$



## Simple Sub-Rayleigh: Strain and Strain Rate III

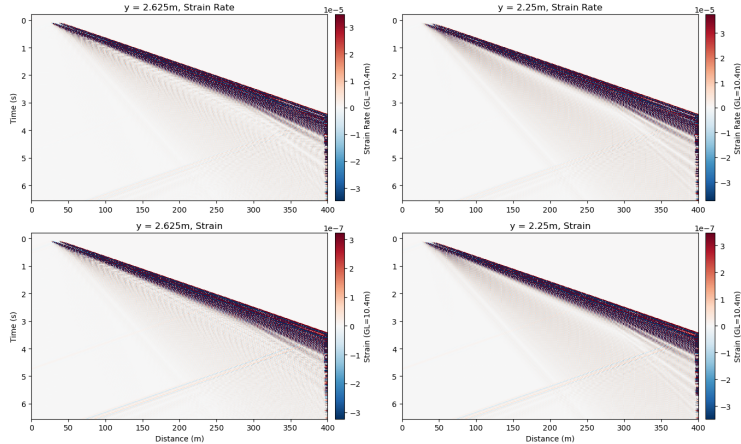
**Figure:** Horizontal Strain and strain rate for sub-Rayleigh crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

- Two-wavefront structure
- Main (thicker) wavefront: slope corresponds to prescribed rupture speed
- Second wavefront: coming only from first source
- Wavefront is emitted by all sources but they interfere (**madariagaDynamicsExpandingCircular1976**), as shown by wider source spacing
- Slope of line close to p-wave speed in medium: S0 wave bounded by plate wave speed limit (**roseWavesPlates2014**)
- S0/P-wave precursor observed in field data by **herwijnenSeismicSensorArray2011**
- Oscillatory behavior of both wavefronts (polarity flip at adjacent timesteps)
- At rupture arrest (270m): v-shaped arrest signature: bidirectional, at fastest speed in medium (**freundDynamicFractureMechanics1998**)

## Simple Supershear: Strain and Strain Rate I

# Simple Supershear: Strain and Strain Rate II

Supershear: Simulated DAS Strain Rate and Strain at  $y = 2.625\text{m}$  and  $y = 2.25\text{m}$





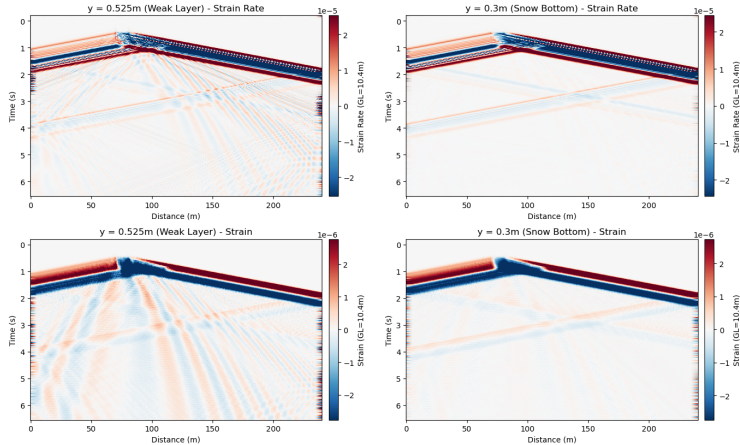
## Simple Supershear: Strain and Strain Rate III

**Figure:** Horizontal Strain and strain rate for Supershear crack propagation in simple two-layer model. Plotted at crack depth (2.625m) and at the snow bottom (2.25m). Gauge length is 10.4m.

- One wavefront, contrary to sub-Rayleigh
- Slope corresponds to rupture speed (within 1m/s of  $v_p$  and  $S_0$ )
- Dispersion and broadening of main wavefront (**kAkiRichards19801981**)
- Oscillatory wavefront
- At  $x=270\text{m}$ , unidirectional fan-like arrest front: rupture outruns  $v_s$ , so arrest energy is only propagated forward (**madariagaHighfrequencyRadiationCrack1977**)
- $S_0$  precursor can be seen in traces

# MPM-Guided: Strain and Strain Rate I

5-Layer Model: Simulated DAS Strain Rate and Strain at  $y = 0.525\text{m}$  and  $y = 0.3\text{m}$



## MPM-Guided: Strain and Strain Rate II

**Figure:** Horizontal Strain and strain rate for MPM-guided model. Plotted at crack depth (0.525) and at the snow bottom (0.3m). Gauge length is 10.4m.

- One two main wavefronts observed
- Crack extent from 75m to 125m: sub-Rayleigh speed from front
- Positive wedge in crack area: extends until rupture arrest, then jumps to same thickness as negative front →quasi-static offset
- Backward-propagating front and forward after crack arrest: close to  $v_p/S_0$  from slope
- Second, weaker wavefront: at onset but also arrest: same arrest structure as sub-Rayleigh
- Oscillatory wavefronts

## Quasi-Static Offset

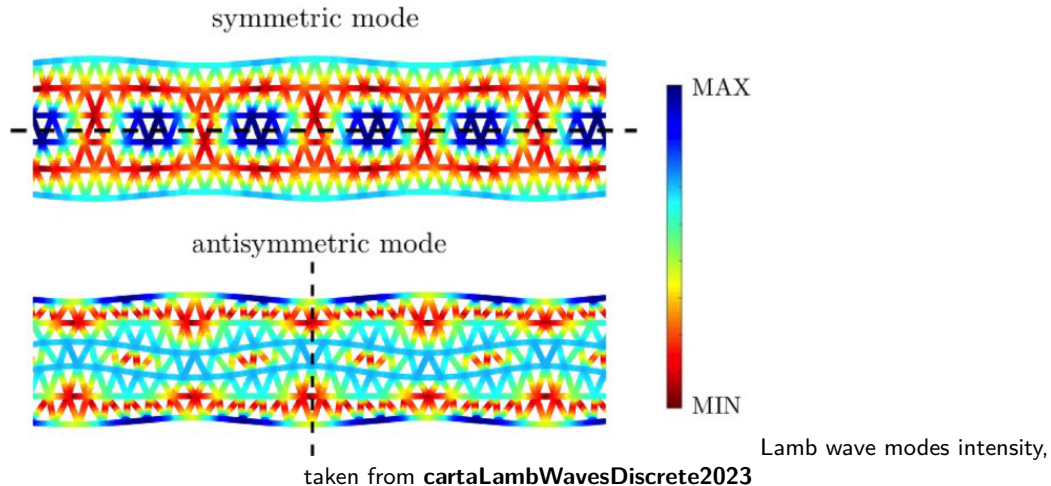
- Static field represents permanent deformation after sources have fired  
(**zhuNoteDynamicStatic2002**; **archambeauGeneralTheoryElastodynamic1968**)
- MPM STFS are not symmetric: contrary to Ricker wavelets, they do not integrate to 0 over time  
→ net displacement of material (**zhuNoteDynamicStatic2002**; **kAkiRichards19801981**)
- Cumulative integral of MPM STFs is negative: collapse of the weak layer, as observed by  
(**heierliAnticracksNewTheory2003**; **heierliAnticrackNucleationTriggering2008**;  
**gaumeDynamicAnticrackPropagation2018**; Trottet et al., 2022)
- All components of MPM STF (directionality) indicate mixed-mode anticrack instead of pure opening or shearing (mode I, II or III)

## Simple model: Lamb Wave Identification I

- Polarity oscillation in all plots: indicative of Lamb waves
- Plotting dispersion curves over f-k spectra: energy is concentrated at symmetric (S0) and antisymmetric (A0) mode
- Antisymmetric mode gets excited preferentially when source is applied normal to midplane of material (**lambWavesElasticPlate1917**; **roseWavesPlates2014**; **achenbachWavePropagationElastic2012**), so in this case yy
- Confirmed by animations of wavefield:

[loop,width=4cm]10../Wave Propagation GIFS/custom<sub>sub</sub>wavefield<sub>2d</sub>moving<sub>v</sub>x<sub>crack</sub>093

## Simple model: Lamb Wave Identification II



# MPM-Guided Model: Lamb Wave Identification I

## Summary of Results & Discussion I



## Implications for DAS field data I

# Outlook

## Answer to Research Questions - Conclusions

# Questions?

# Bibliography I



Bergfeld, B., Gaume, J., Bobillier, G., Pellet, A., Schweizer, J., & van Herwijnen, A. (2025). Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments. *Nature Communications*, 16(1), 11153.  
<https://doi.org/10.1038/s41467-025-65825-6>



Edme, P., Paitz, P., Walter, F., van Herwijnen, A., & Fichtner, A. (2023, February). Fiber-optic detection of snow avalanches using telecommunication infrastructure.  
<https://doi.org/10.48550/arXiv.2302.12649>



Gaume, J., van Herwijnen, A., Chambon, G., Birkeland, K. W., & Schweizer, J. (2015). Modeling of crack propagation in weak snowpack layers using the discrete element method. *The Cryosphere*, 9(5), 1915–1932.  
<https://doi.org/10.5194/tc-9-1915-2015>

## Bibliography II



Gaume, J., van Herwijnen, A., Chambon, G., Wever, N., & Schweizer, J. (2017). Snow fracture in relation to slab avalanche release: Critical state for the onset of crack propagation. *The Cryosphere*, 11(1), 217–228.  
<https://doi.org/10.5194/tc-11-217-2017>



Paitz, P., Lindner, N., Edme, P., Huguenin, P., Hohl, M., Sovilla, B., Walter, F., & Fichtner, A. (2023). Phenomenology of Avalanche Recordings From Distributed Acoustic Sensing. *Journal of Geophysical Research: Earth Surface*, 128(5), e2022JF007011. <https://doi.org/10.1029/2022JF007011>



Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, 41(4). <https://doi.org/10.1029/2002RG000123>



SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.

## Bibliography III

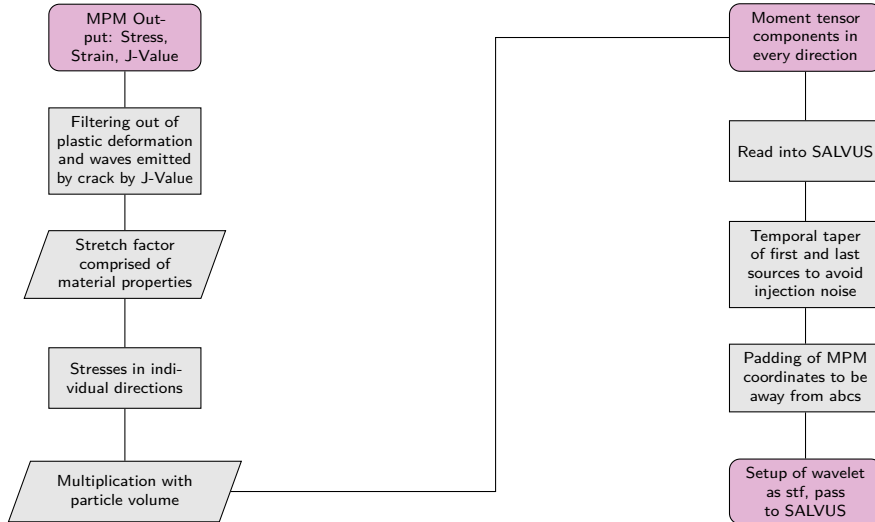


Trottet, B., Simenhois, R., Bobillier, G., Bergfeld, B., van Herwijnen, A., Jiang, C., & Gaume, J. (2022). Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches. *Nature Physics*, 18(9), 1094–1098.  
<https://doi.org/10.1038/s41567-022-01662-4>



Turquet, A., Wuestefeld, A., Svendsen, G. K., Nyhammer, F. K., Nilsen, E. L., Persson, A. P.-O., & Refsum, V. (2024). Automated Snow Avalanche Monitoring and Alert System Using Distributed Acoustic Sensing in Norway. *GeoHazards*, 5(4), 1326–1345. <https://doi.org/10.3390/geohazards5040063>

# MPM STF Workflow







Salome Bachmann  
Master Student  
sbachmann@student.ethz.ch

ETH Zurich  
D-EAPS